



Exploring the impact of computer-mediated emotional interactions on human facial and physiological responses

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ABSTRACT

Remote communication has become pervasive, yet its impact on human emotions, empathy, and physiological responses remains unclear. This study addresses this gap by investigating how emotional video-mediated conversations differ from face-to-face interactions, focusing on behavioral and physiological responses. We create a multimodal dataset of Electrodermal Activity (EDA) signals, Photoplethysmography (PPG) signals, and facial videos from two people conversing about emotional topics in face-to-face and remote video-mediated conditions. We use a series of repeated measures ANOVA with aligned rank transform to compare face-to-face to remote conversation in terms of heart rate activity, electrodermal activity, and facial action units. We also explore how subjective empathy between people varies in these two conditions. Our findings reveal significant differences in physiological responses between face-to-face and remote conversation and variations in perceived empathy based on interaction setting, highlighting the nuanced influence of communication channels. We also show that we can recognize emotions more accurately when we pre-train a random forest classifier with one condition's data (an increase of 20% to 45% for various modalities). Finally, we discuss the research findings and limitations and offer insights for optimizing human-computer interaction and understanding human emotional responses in an increasingly tech-mediated world.

1. Introduction

During the COVID-19 pandemic, people often interacted remotely with colleagues, health professionals, family, and friends through video-conferencing [1] and other online tools. However, in this situation, people experience different environments, there is a lack of some social cues (like whole-body gestures), and they can be unfamiliar with the technology being used [1,2]. The result is that people may communicate less and feel emotions differently than in a face-to-face setting.

Understanding other people's emotions and having empathy for others is an essential skill in human-human communication. Empathy is the ability to put yourself in others' place [3] and observe, understand, and share another person's feelings [4]. Empathy can be comprised of affective, cognitive, and somatic components; Affective empathy involves sharing one's feelings with others, Cognitive empathy

is the cognitive ability to comprehend others' perspectives, Somatic empathy is a person's physiological response to another's emotion. In social relationships, empathy is crucial because it enables people to understand each other's emotional states [5]. It has been shown that being able to show empathy reduces emotional and physiological distress [6] and improves relationships [7].

Previous studies have shown that our body and brain activity can influence another person's body and brain activity [2,8]. For example, the breathing pattern of the patient and therapist can become synchronized when they are close to each other [9]. Golland et al. [10] showed convergence in physiological and emotional states among co-present people while doing a task. This helps people to regulate their emotions. A key question is whether this is the same for people when communicating remotely. Rolink and Ehrenreich [2] discuss the lack

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Table 1
An overview of multimodal datasets with physiological data.

Dataset	Modalities	Stimuli	Size
DEAP [11]	ECG, EDA, EEG, EMG, EOG, RESP, SKT, face video	40 music videos	32
MAHNOB-HCI [12]	ECG, EDA, EEG, RESP, SKT, face and body video, eye gaze tracker, audio	20 movie clips	27
RECOLA [13]	video, audio, EDA, and ECG	NASA survival task	46
SEED_IV [14]	EEG data and eye movement	72 Chinese film clips	15
AMIGOS [15]	EEG signal, ECG signal, GSR signal, frontal full body video and depth video	16 short videos alone, four long videos in group	40
PEGCONV [16]	EEG, PPG, EDA, video, audio	face-to-face conversations	23

of important information like body gestures, smells, and eye gaze in remote therapy, which may impact remote communication.

Although the benefits of remote communication is clear, there needs to be more knowledge about its effect on human emotions, empathy with others, and, consequently, human physiological responses. We need to better understand the impact of video conferencing on human communication compared to face-to-face communications to develop better remote interaction systems.

Computer interfaces have often been improved through emotion recognition, for example, by adjusting sounds, user interfaces, graphics, and content based on user emotion [17]. Emotions are multimodal processes related to neural, physiological, and behavioral activities such as body movements and facial expressions, brain activities, heart rate variability, and galvanic skin response [18]. Many studies have shown the effectiveness of using these internal and external body responses for recognizing emotions [19,20]. Previous studies have shown that combining different physiological and behavioral modalities improves emotion recognition accuracy [14,18]. Saffaryazdi et al. [16] examined various behavioral and physiological cues, particularly in face-to-face conversations, and demonstrated the effectiveness of various cues and their combinations in emotion recognition. We are curious about whether this is the same for video-mediated discussions. We also want to investigate whether trained models in face-to-face conversations can be used to recognize emotions in remote conversations. This could lead to the development of technology that enables people to better connect emotionally over distance.

This paper aims to compare human emotional responses in different conversational settings. To achieve this aim, we investigated three research questions (RQs):

- RQ1: Is there any difference between human physiological responses in face-to-face and remote communications in emotional contexts?
- RQ2: Do people experience the same level of empathy in face-to-face communication compared to video-mediated communication?
- RQ3: Can we use trained emotion recognition models from face-to-face conversations for predicting emotions in remote video-mediated conversations?

In this paper, we compare the effect of remote emotional conversations on human heart rate, electrodermal activity, and facial expressions and compare it with human responses in face-to-face conversations. We statistically show what features and aspects of physiological and facial activities vary in different emotional and conversational conditions. We also explore how empathy between people varies subjectively in remote and face-to-face communication. The main contributions of this paper are as follows:

- Exploring the effect of video-conferencing on physiological and behavioral responses in emotional conversations.

- Identifying the differences in human body responses between face-to-face conversations and remote conversations.
- Identifying the most effective way to recognize emotions in different settings.
- Creating a multimodal dataset of EDA, PPG, and facial video data captured from face-to-face and remote conversations and making it publicly available.¹

2. Related work

The COVID-19 pandemic prompted a widespread transition to remote communication methods. Organizations, families, and individuals rapidly adopted video-conferencing and other online tools to facilitate interactions and maintain connectivity while adhering to social distancing measures [1]. The advantages of video conferencing and remote calls are clear. According to Nguyen et al. [21], remote communication is critical when people are physically apart or when face-to-face communication is impossible. For example, Sorlie et al. [22] showed that video-conferencing is a reliable option in psychotherapy when there is a lack of face-to-face supervision available.

Hundreds of studies have been conducted over the last 60 years to qualitatively compare remotely with face-to-face conversations and the effects of technology-mediated communication on humans. For example, Knipe and Lee [23] found that the quality of teaching and learning in online classes is less than in face-to-face classes. Similarly, Serhan et al. [1] discussed some reasons for poor communication in remote conversations, including environmental distractions, lack of physical connection, and technical issues. Bailenson et al. [24] write about the psychological consequences and fatigue caused by remote communications. They suggested more research is needed to make remote communication as effective as face-to-face communication.

Prior research has discussed the potential differences in emotional experience and communication effectiveness between remote and face-to-face settings. Krzyzaniak et al. [25] systematically reviewed the effectiveness of telehealth versus face-to-face interventions for anxiety disorders. They discussed telehealth interventions for treating anxiety and related conditions that have shown comparable effectiveness to in-person therapy. Yet, additional high-quality trials are needed to fully assess the efficacy, acceptability, feasibility, and cost-effectiveness of telehealth interventions across a broader spectrum of anxiety disorders and treatments.

Rolnick and Ehrenreich [2] discussed the limitations of online video psychotherapy, like missing vital information such as posture, sitting, breathing, smells, and various motor movements. Ehrenreich et al. [26] discussed the advantages of using and sharing biofeedback, like heart rate or skin temperature, in psychotherapy. For example, patients can find new ways of minding the body by learning psychophysiological patterns. Irvine et al. [27] reviewed the telephone therapy literature

¹ <https://pegconv.nastaran-saffar.me/>

and identified the need for enhanced or augmented listening by the therapist to detect affect and emotion accurately. However, many researchers found that the absence of visual co-presence can be helpful to some patients by offering greater anonymity and reducing anxieties around visiting a clinic setting or identifying social status [27–29]. Irvine et al. [27] identified only a limited group of studies examining differences in interactional dynamics between telephone and face-to-face therapies, thereby constraining the certainty of any definitive conclusions.

Nowadays, detecting emotions, especially using multimodal input, has attracted a significant amount of research because of its crucial role in enhancing computer interfaces [30,31]. Sharma et al. [31] have provided a comprehensive list of multimodal datasets, but only a limited number collected physiological signals from people in response to stimuli. Table 1 shows some important details of multimodal datasets with both behavioral and physiological inputs. As shown in Table 1, visual stimuli were commonly used to induce emotions in individuals, and only two of them used interactive tasks between people. Although these datasets are very useful in training emotion recognition models, to the best of our knowledge, no research has explored whether emotional responses are the same in face-to-face and remote interactions. It is also interesting to explore if trained models on emotional responses in face-to-face conversations can be used for recognizing emotions in remote conversations.

As mentioned in this section, many researchers qualitatively explored the effect of remote communications on humans [21,32]. However, little research has been conducted on how remote communications affect human physiology and emotions, and how physiological responses and facial movements may differ in remote communication compared to face-to-face, particularly in an emotional context. Also, previous research created multimodal datasets and explored multimodal emotion recognition, mostly in the presence of visual stimuli or face-to-face conversation. To the best of our knowledge, there is no comparative multimodal dataset of human body responses in face-to-face and remote conversations.

Addressing the identified gap in the effect of remote communication on humans, this research creates a dataset containing physiological signals and behavioral responses collected during emotional conversations in both settings. We aim to answer questions like: Is there any difference between human responses in face-to-face and remote communication in emotional contexts? Do people experience the same level of empathy in face-to-face communication compared to video-mediated communication? Can we use trained emotion recognition models from face-to-face conversations for predicting emotions in remote video-mediated conversations? By creating a comparative multimodal dataset from emotional conversations and then using statistical analysis and machine learning methods, we aim to offer deeper insights into designing more effective remote interaction tools and understanding the impact of remote interaction on human physiological responses and emotional expression.

3. Preliminaries

3.1. Emotion representation

Researchers represent emotions in two main ways: discrete and dimensional emotion models. Ekman's basic emotion model is a well-known discrete emotion model that categorizes emotions into six basic types: happiness, sadness, surprise, anger, disgust, and fear [33]. In contrast, the dimensional emotion model defines emotion as a combination of three components: arousal, valence, and dominance [19]. Valence represents a range of negative to positive emotions. Arousal represents a range of passive to active emotions. In this study, we used Russel's Circumplex Model [34], a dimensional model that only uses valence and arousal to represent emotions.

3.2. Emotion induction methods

In order to study the impact of video conferencing on emotional conversations, we must naturally induce emotions. This challenging task can be done in different ways, including visual stimuli like watching images or videos, listening to music, recalling personal emotional memories, accomplishing psychological or physiological procedures, imagining emotional scenes, and using virtual reality [35,36]. Siedlecka et al. [35] empirically showed that recalling and imagining memories is one of the most robust ways of inducing emotions. In this study, we used a combination of visual stimuli, memory recall, and talking about past emotional experiences to create an emotional conversation.

4. Methodology

The nature of the research questions demands a quantitative approach since they involve the measurement and comparison of physiological and behavioral indicators, such as heart rate variability (HRV), electrodermal activity (EDA), and facial expressions across face-to-face and remote conversations. So, we used a quantitative approach to explore the effect of remote vs. face-to-face conversational settings on human physiology, behavior, and various empathy factors. A within-group experimental study was conducted to collect the various physiological and behavioral cues, and we used statistical analysis to explore significant differences in human body responses in these two settings. We also compared participants' empathy levels in a face-to-face and remote setting due to emotional changes. We considered the type of conversation and the reported arousal and valence levels as the independent variables and various features of electrodermal activity (EDA), photoplethysmography (PPG), facial action units, and reported empathy factors as the dependent variables.

4.1. Study design

A total of 15 participants (7 females and eight males) aged 21–36 years old were recruited from the university ($M=28.6$, $SD=5.7$). This sample size falls within the typical range of 15–32 participants, observed in well-known multimodal datasets involving physiological data for emotion recognition like DEAP [11], MAHNOB-HCI [12], and SEED-IV [14]; see Table 1). To compare human emotional responses in face-to-face conversation with remote conversation, we needed similar emotional conversations in two conditions that covered all of the emotional responses. For this reason, we created identical emotional conversations regarding length and content in face-to-face and remote conditions. We used a similar strategy described in Saffaryazdi et al. [16] to stimulate emotions in conversations. In this study, each participant had a set of conversations with an expert about their emotional memories and experiences. Instead of using previous memories and experiences about each emotion as the topic of conversations, we used previous memories and experiences related to a collection of IAPS images [37] to keep the content for all participants similar. In each trial, participants conversed with a female expert about their related memories and experiences triggered by the IAPS image.

We used 16 images from the IAPS dataset as the topic of conversation. We sorted the IAPS images based on reported arousal and valence levels and selected four images from each of the following categories: High Arousal-High Valence (HAHV), High Arousal-Low Valence (HALV), Low Arousal-High Valence (LAHV) and Low Arousal-Low Valence (LALV). The images were selected based on subjects that are easy to discuss. Due to ethical concerns, we ignored images that dealt with negative topics such as suicide, violence, addiction, and so forth. A table of selected images and their details is shown in Appendix A.

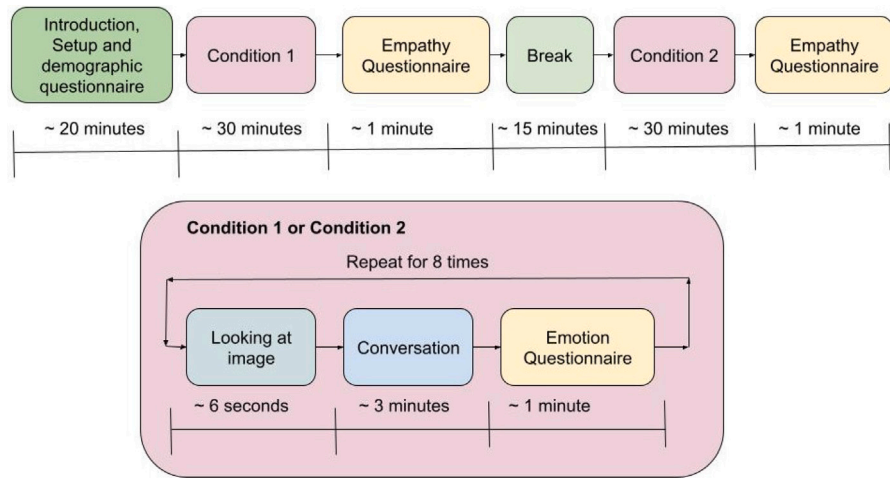


Fig. 1. The overall procedure of the user study.

4.2. Procedure

The experiment began by explaining the goals, devices, and steps, and completing a consent form and demographic questionnaire. All participants had both face-to-face and remote conversations, alternating the start condition and the topic (image set) in a counterbalanced way. In the face-to-face condition, the participant sat about one meter from the interviewer while an image was displayed on a monitor nearby. In the remote condition, the interviewer was in another room and shared the screen of the image with the participant. Half of the screen was the interviewer's upper body and face, and half was the image. Each image remained on the screen during the conversation. The order of pictures in each condition and between the two conditions was chosen randomly. In each condition, we displayed two images from the four categories of LVLA, LVHA, HVLA, and HVHA.

After setting up the data acquisition devices (explained below), the interviewer entered the experiment room or started a Zoom video conferencing call from another room. First, the interviewer had a friendly conversation with the participant to establish a friendly atmosphere. The two conditions were then performed consecutively with a 10-minute break in between.

Each conversation began by displaying a fixation cross (3 s) on the monitor, followed by displaying a randomly selected image of our IAPS image set (6 s). The interviewer then discussed with the participant their feelings about the image, similar experiences, or general discussion about the image for two to three minutes. The detailed questions that the interviewer used to have the conversation can be found in [Appendix B](#).

After each conversation, we used a questionnaire based on the self-assessment manikins (SAM) questionnaire [38]. We asked participants to rate their arousal and valence level after each conversation topic from 1 to 9. [Fig. 1](#) shows the steps of the experiment procedure.

Participants also completed a self-report questionnaire, reporting their overall empathy toward the interviewer during face-to-face or remote conditions. This questionnaire was made up of the following questions:

- I felt the interviewer was empathetic toward me (rate 1 to 10) [39].
- I felt empathetic toward the interviewer (rate 1 to 10) [40].
- The interviewer identified my feelings (Cognitive empathy, rate 1 to 5).
- The interviewer reflected on my emotions (Affective empathy, rate 1 to 5).
- I could interact with the interviewer easily (rate 1 to 5).

The first four questions were created according to the definition of empathy and previous studies to evaluate empathy [39,40]. We also added one question to compare the ease of communication in these conditions.

4.3. Data acquisition

We used an OpenBCI EEG cap, Shimmer3 EDA+, the Intel RealSense camera, and a Logitech webcam to capture EEG, PPG, EDA, facial video, and audio data, respectively. Unfortunately, we had to exclude EEG data from the data analysis because the OpenBCI headset failed to stream data for most of the participants in at least one of their conditions. We used the Octopus-sensing software suite [41] to acquire data synchronously. [Fig. 2](#) shows the experiment setup and the devices we used.

The experiment was conducted using three laptops. One was used to run the experiment procedure, connect to the sensors, and record data. On the second laptop, participants filled out questionnaires or made Zoom calls. The interviewer used a third laptop in remote conditions to make Zoom calls and control the procedure. In remote conditions, the first laptop recorded data, like in the face-to-face one. The Octopus-Sensing library was used to manage the experiment procedure over the network. To make Zoom calls, we used the second laptop's microphone, speaker, and webcam, but to record data, we employed the same audio recorder and camera as in the face-to-face condition.

4.4. Data coding

Participants rated their arousal and valence levels from 1 to 9, where 1 meant a calm state for arousal, a negative state for valence, and 9 meant an excited state for arousal and a positive state for valence. Using the middle rate as the threshold, we coded arousal and valence levels with a 0 (low) or 1 (high), respectively. We also coded face-to-face conversations as 0 and remote ones as 1.

5. Data analysis

We collected video, PPG, GSR, and EEG signals from 15 participants, and each had eight trials in each condition. However, the EEG data was not valid for most participants because the OpenBCI sensor failed in data streaming at least in one condition. So, we excluded EEG data from data analysis. In order to prepare each modality for statistical analysis and emotion recognition, we applied some preprocessing steps. We extracted several features from the captured data described in [Table 2](#). Based on the extracted features from preprocessing, several 3-way repeated measures ANOVA with aligned rank transform (ART) [42] were used to examine how physiological and behavioral responses vary in face-to-face and remote settings.

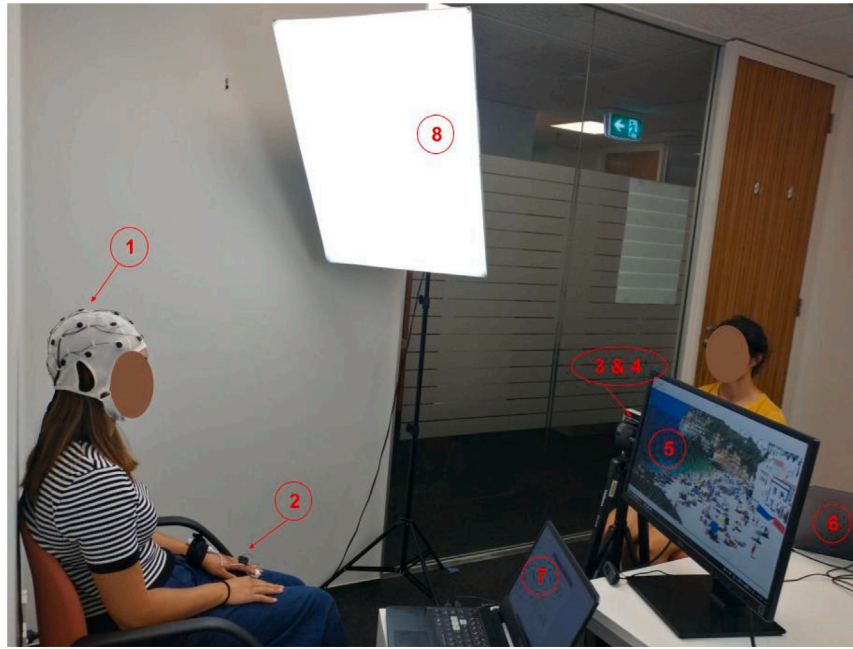


Fig. 2. Experiment setup with the participant and interviewer in a face-to-face conversation. 1-EEG cap, 2-Shimmer3 sensor, 3&4-camera and microphone, 5-The monitor for displaying images and the interviewer's upper body in the remote condition, 6-main laptop, 7-second laptop, 8-lighting softbox.

5.1. Preprocessing and feature extraction

We first extracted only the data recorded during the conversations to prepare the data for statistical analysis. For this reason, we removed the first 9 s of data related to watching the fixation cross and the image. Then, we considered the first 3 s of these 9 s as baseline data.

5.1.1. Facial video

We used the OpenFace toolkit [43] to identify and track 18 facial action units² which are related to emotional expressions. Then, for each action unit, we used a majority vote method among all video frames to measure the presence of an action unit in each video.

5.1.2. EDA signal

We used the neurokit2 library [44] to clean the EDA signal and extract EDA peaks and phasic and tonic components. Then, we calculated the mean, median, standard deviation, variance, maximum, minimum, the sum of positive derivatives, and the sum of negative derivatives of the extracted components. Table 2 shows the extracted features from the EDA signal. Finally, we normalized the features of each participant's trials and transferred them to the [0,1] range.

5.1.3. PPG signal

To prepare the PPG signal for feature extraction, first, we used neurokit2 to clean the signal, then we applied a bandpass filter in the range of 0.5 Hz to 4 Hz using the heartpy library [45] to filter the signal. Finally, we applied baseline normalization and subtracted the baseline average from the PPG signal. We used the neurokit2 library to extract time domain features of HRV signals. Table 2 shows the extracted features from the PPG signal. Finally, we normalized each participant's trials similar to EDA signals.

5.2. Statistical analysis methods

We performed a within-group statistical analysis as all participants conversed face-to-face and remotely. Our study consisted of three independent variables: condition (face-to-face and remote), arousal level, and valence level. Since our data was not normally distributed, we used the aligned rank transform (ART) with repeated measures ANOVA [42], a non-parametric method allowing multiple independent variables, interactions, and repeated measures.

In this experiment, we had more than one sample for each low and high arousal and valence level. So, we measured the average of samples with the same arousal or same valence levels and then applied the repeated measures ANOVA with ART. We analyzed data using Python and the ARtool package [46] in R. A series of three-way repeated measures ANOVA with ART was conducted to evaluate the effect of condition (face-to-face or remote), emotional arousal, and emotional valence on human facial expressions, electrodermal activity, heart rate, and heart rate variability. We also applied two-way repeated measures ANOVA with ART to evaluate the effect of condition and gender on different empathy measurements.

5.3. Emotion recognition

In this study, we were interested to know if trained models in one setting could accurately recognize emotions in another setting. For this reason, we used the Random Forest Classifier (RFC), previously used in emotion classification research [47,48] to classify data for high and low arousal and valence. We used random search cross-validation for tuning RF estimators and found that using 200 estimators, we got the best result. We used the default values from the Sklearn library,³ and the extracted features in the previous section as the feature vectors. We split data into 20-second windows to increase the number of samples and made all samples the same length. We did not consider windows smaller than 20 s because this leads to inadequate data for the heartpy library to extract heart rate features.

² <https://www.cs.cmu.edu/~face/facs.htm>

³ <https://scikit-learn.org/stable/index.html>

Table 2

The extracted features from the recorded data. *Statistics features are mean, median, maximum, minimum, variance, standard deviation, the sum of positive derivatives, and the sum of negative derivatives.

Data	Features
EDA	phasic statistics, tonic, and peaks statistics
PPG	Heart rate variability time domain features
Face	OpenFace action units including these numbers: 1, 2, 4, 5, 6, 7, 9, 10, 12, 14, 15, 17, 20, 23, 25, 26, 28, and 45

Table 3

Facial Action units that are significantly affected by each factor. A:V (Arousal:Valence).

Factor	Feature	Description	F	Df	Df.res	P-value
Condition	AU20	Lip stretcher	33.46	1	72.25	< 0.01
	AU06	Cheek Raiser	26.77	1	72.08	<0.01
	AU14	Dimpler	18.97	1	72.04	<0.01
	AU45	Blink	18.54	1	72.27	< 0.01
	AU05	Upper Lid Raiser	14.04	1	72.15	<0.01
	AU12	Lip Corner Puller	11.92	1	72.03	<0.01
	AU09	Nose Wrinkler	8.60	1	72.10	<0.01
	AU25	Lips part	7.46	1	72.30	<0.01
	AU04	Brow Lowerer	6.37	1	72.18	0.01
	AU15	Lip Corner Dep	4.57	1	72.45	0.04
Valence	AU06	Cheek Raiser	12.06	1	72.22	<0.01
	AU12	Lip Corner Puller	7.77	1	72.07	0.01
	AU10	Upper Lip Raiser	7.40	1	72.06	0.01
	AU04	Brow Lowerer	5.45	1	72.42	0.02
	AU25	Lips part	4.88	1	72.74	0.03
	AU45	Blink	4.23	1	72.66	0.04
Arousal	AU10	Upper Lip Raiser	4.75	1	72.04	0.03
	AU12	Lip Corner Puller	3.98	1	72.04	0.05
A:V	AU14	Dimpler	4.27	1	72.03	0.04

For the PPG and EDA data, we extracted the features of each window and made a feature vector. We assigned the label of each trial to its parts. For the facial video, we extracted the action units of each frame, measured the sum of action unit intensities and the number of frames with each action unit, and made a feature vector for every 20 s of data.

6. Results

We applied a three-way repeated measures ANOVA with ART for each extracted feature with the condition, arousal level, and valence level as the three factors. Only a few features of recorded data significantly differed in various conversation settings and emotional conditions. Each extracted feature revealed one aspect of the data; only some aspects differed in various conditions. We also plotted the point plot of significant features using the Seaborn library [49] in Python. Each point plot estimates the central tendency of an essential feature according to a categorical variable like conversation condition, arousal, or valence level and shows the relationship between levels of the categorical variable with the notable features.

6.1. Facial activity

The condition significantly impacted the majority of extracted action units pertaining to various facial regions (10 out of 18), with p -values less than 0.04. Significant features affected by each factor are presented in Table 3. To compare facial expressions between face-to-face and remote conversations, we specifically plotted AU20 and AU45, which were more significantly different. Similar patterns were observed in the point plots of AU05, AU09, AU25, and AU15, resembling AU20, while AU06, AU12, and AU14 demonstrated similarities to AU45.

In Fig. 3, the point plot displayed a higher average frequency of AU20 presence, along with AU05, AU09, A15, and A25, during face-to-face conversations. These action units are primarily associated with

negative emotions. Conversely, the presence of AU45, AU06, AU12, and AU14, which relate predominantly to positive emotions, exhibited a higher average tendency in remote conditions. Thus, based on facial expressions, individuals appear to experience more positive emotions during remote conversations than in face-to-face conversations.

Furthermore, specific muscles around the lips and eyes indicated distinct action units depending on valence levels (p -value < 0.04). Point plots for the presence of AU25, AU06, AU10, and AU12 were analogous in valence level and lower during instances of low valence. These action units are primarily associated with positive emotions, which are more prevalent in positive conversations. Similarly, the presence of AU45 and AU04 followed a parallel pattern, with higher occurrences in low valence. AU04 is associated with facial expressions related to anger, fear, and sadness, which were more frequent during negative conversations. Additionally, the blinking frequency (AU45) was elevated in negative conversations, as depicted in Fig. 5.

Regarding arousal levels, only the upper lip raiser and lip corner puller were significantly affected (p -value < 0.05). As shown in Fig. 4, the presence of AU12 and, similarly, AU10, which are linked to upper lip movements and contribute to expressions of happiness, exhibited lower occurrence during instances of low emotional arousal.

6.2. EDA signal

The average of the phasic component was the only feature of the EDA signal that was significantly affected by the conversation condition (p -value = 0.026). In face-to-face conversations, the average tendency of phasic mean was considerably higher than in remote conditions (Fig. 3, third row). This shows on average, participants' reactions to stimuli were more substantial and immediate in face-to-face conversations.

The valence level significantly affected EDA peaks' frequency (p -value = 0.037), peaks' positive derivatives (p -value = 0.037), and peaks' variance (p -value = 0.049). The correlation analysis showed these features are highly correlated. Their point plot was quite similar, so we plotted only one. Fig. 5 (third row) shows that the average variance of EDA peaks and similarly peaks' frequency and peaks' positive derivatives in negative emotional states is higher than in positive emotional states. When the EDA signal exhibits more frequent peaks, it suggests that there are more instances of rapid increases in skin conductance followed by decreases because of changes in emotional state.

Regarding the arousal level, the minimum of the tonic component was significantly affected by arousal level, which was considerably higher at low emotional arousal compared to high emotional arousal. Several features were significantly affected by the interaction effect of two or three factors. Table 4 shows the features affected considerably by each factor and their interactions. This result is in contrast with the definition of the tonic component. When the minimum value of the EDA tonic component is lower, it indicates that the skin is exhibiting less electrical conductance at its baseline state. This could be interpreted as a lower overall arousal or physiological activation level. However, it is essential to recognize that the minimum of the tonic component might not capture the comprehensive pattern of the EDA signal, and regulatory mechanisms might contribute to its decrease.

Table 4

EDA features significantly affected by each factor. C: Condition, V: Valence, A: Arousal.

Factor	Feature	F	Df	Df.res	P-value
Condition	phasic_mean	5.11	1	83.70	0.026
Valence	peaks_pos_dv	4.51	1	86.10	0.037
	peaks_freq	4.49	1	85.14	0.037
	peaks_var	3.98	1	85.56	0.049
Arousal	tonic_min	4.24	1	87.49	0.042
C:V	max	4.07	1	84.54	0.047
	phasic_mean	4.37	1	84.11	0.040
C:A	phasic_median	4.01	1	83.68	0.049
A:V	phasic_median	5.29	1	83.68	0.024

Table 5

PPG features that are significantly affected by each factor.

Factor	Feature	F	Df	Df.res	P-value
Condition	HRV_MadNN	7.95	1	84.51	0.006
	HRV_MCVNN	6.87	1	84.51	0.01
	hrv_madnn	5.38	1	83.29	0.023
	HRV_IQRNN	4.71	1	84.4	0.033
Valence	HRV_TINN	4.25	1	84.13	0.042

6.3. PPG signal

The nature of the conversation had a significant impact on certain aspects of the derived PPG signal features. The p-values of significantly different PPG features are presented in Table 5. In Fig. 3, HRV-MadNN, HRV-MCVNN, and HRV-IQRNN exhibited a significantly higher value during remote conversations. The elevation in these features indicates increased heart rate variability, which has been associated with improved emotional well-being [50]. Furthermore, higher HRV has been linked to decreased stress levels, reduced anxiety [51], and enhanced regulation of emotional reactions [52].

This result could be because the interviewer was a stranger to participants, and having face-to-face conversations with strangers for the first time can be stressful. Additionally, face-to-face conversations can be more difficult to escape if they become uncomfortable. In face-to-face conditions, people have a higher sense of presence than in remote conditions. So they may get more nervous talking to other people. People in a remote conversation and on a screen do not have the same spatial cues as face-to-face people and so do not feel so real. Consequently, our findings suggest that engaging in remote conversations enabled participants to have better emotional regulation. Furthermore, higher HRV is linked to increased attention [52], implying that participants were potentially more effectively engaged in conversations during the remote condition. This may be because people feel more stressed in face-to-face interaction with strangers. In comparison, they feel more comfortable in remote conversations because they used to have remote meetings with strangers daily because of their jobs.

Only the average value of HRV-TINN was significantly different at various valence levels. It was significantly higher in low valence levels. A higher HRV-TINN value indicates more significant variability in the timing of heartbeats, which is often associated with healthier autonomic nervous system function and reduced stress [53], possibly reduced positive stress in this experiment. Also, since HRV decreases with increasing task demands [54], a lower level of HRV in positive conversations may indicate more engagement in positive conversations.

We observed no significant differences between PPG features at the different arousal levels. However, based on the literature, heart rate variability indicates arousal level. A potential explanation for this outcome is that the conversations were ineffective in inducing various arousal levels.

6.4. Empathy

Using the empathy questionnaire, we examined four empathy factors. We used a one-way repeated measures ANOVA with ART. Only the participants' feelings about the interviewer's empathy toward them significantly differed between the two conversation settings (p -value = 0.04). While the interviewer was the same in both conditions, remote participants felt less empathy from the interviewer, possibly because of the reduced sense of social presence. Other empathy factors were not significantly different between the two conditions.

In addition, we plotted points to identify the relationship between conversation settings, gender, and each empathy factor, as shown in Fig. 6. Different genders showed various levels of empathy toward the interviewer in the two conversation conditions. However, they had no significant difference (p -value = 0.29). According to these graphs, some factors, including reflection and interviewer empathy toward participants, decreased in remote conversations but were insignificant.

There was also no significant difference between the two conversation settings according to the interaction measures. This means participants had no difficulty interacting with the interviewer in remote interaction. However, the point plot shows that face-to-face interaction between the interviewer and participant was easier.

6.5. Emotion recognition

We used the leave-one-subject-out (LOSO) strategy, also known as the leave-one-out cross-validation (LOOCV) strategy, to evaluate our methods for emotion recognition. At first, we used a 5-fold evaluation strategy to classify data from scratch in face-to-face and remote conditions. Then, we considered the trained model with the highest accuracy as the base model. We used the base model in one condition to initialize the model for another condition. We also used the base model in each condition to recognize arousal and valence in the other condition.

Table 6 compares these methods in recognizing arousal and valence levels. In the Kfold method, the RFC model starts from scratch, and the accuracy is the best accuracy of 5-fold validation when training and test data are in one condition (f2f or remote condition). In the LOSO method, the RFC model also starts from scratch, and the accuracy is the average accuracy of leave-one-subject-out when training and test subjects are in one condition. The subject-out data is unseen data, so accuracy is low. In the Pre-trained method, the model is initialized with the fitted model on another condition's data. In this method, we use the best model of k-fold fitted in one condition to initialize the model for the other condition. LOSO evaluation tests the model on an unseen participant's trials. However, since we used the other condition's trained model for initializing the model, the model has seen subject-out trials in another condition. So, it works like a leave-some-trial-out evaluation. The high accuracy of this model shows the high similarity of physiological responses in these two conditions. The subject-out data are unseen data. However, the model has seen subject-out data in another condition and trained on other subjects in the same condition with subject-out. So, the model has learned about the subject-out but in different conditions and about the other subjects in the same condition. We also trained the RFC model on one condition's data from scratch and used another condition's data as the test set. The test subjects' data are unseen data. However, the model has seen trials of all subjects in another condition.

The reported accuracies in Table 6 are the average of LOSO evaluations for all subjects. As can be seen, a pre-trained model with one condition's data accurately recognizes arousal and valence levels for another condition. This means that the data from these two settings are consistent enough, which led to this high accuracy. Furthermore, according to this table, the accuracy of trained models in one setting is close to that of the trained model with each setting's data when the evaluation method is LOSO.

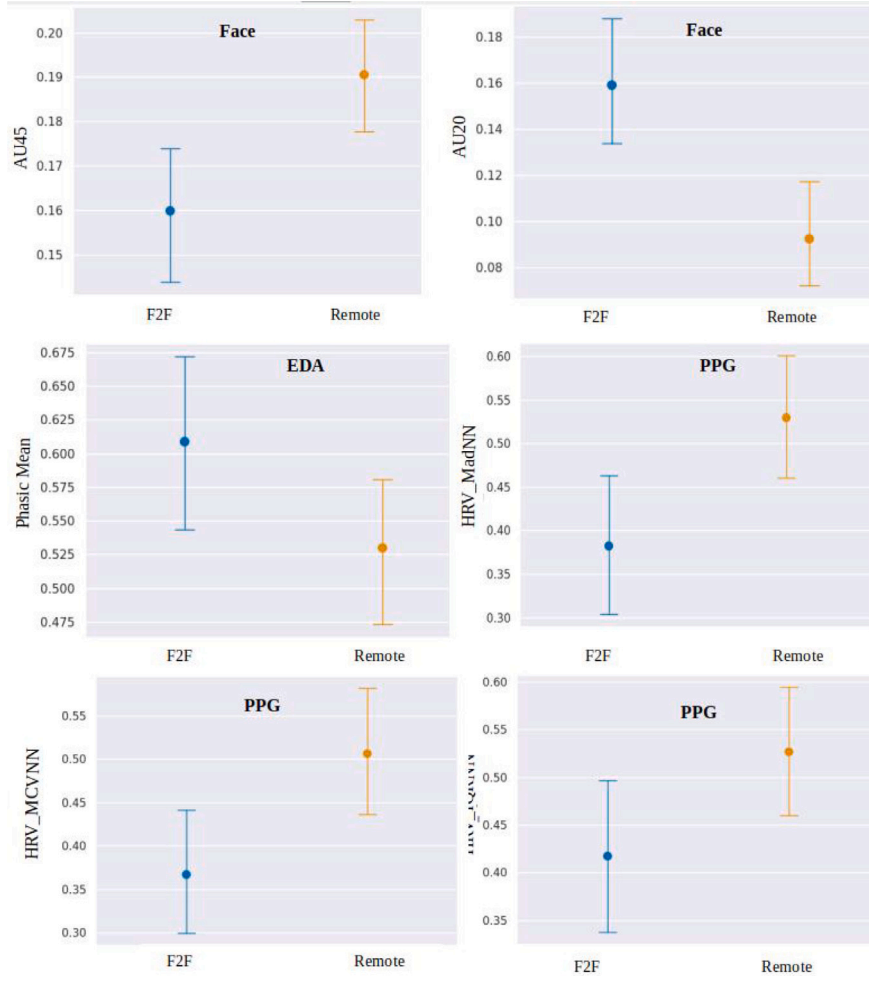


Fig. 3. The point-plot of features significantly affected by conversation condition. For PPG signal, standard deviation, peaks' mean, peaks' min, peaks' var, peaks' std, peaks' positive derivatives, peak's median, peak's max, peaks' negative derivatives, max, and median are highly correlated with similar point plot, so we only plotted standard deviation. For the face, the plot of AU20, AU05, AU09, AU25, AU15 were similar to AU20 and AU14, AU06, AU12 and AU04 were similar to AU45.

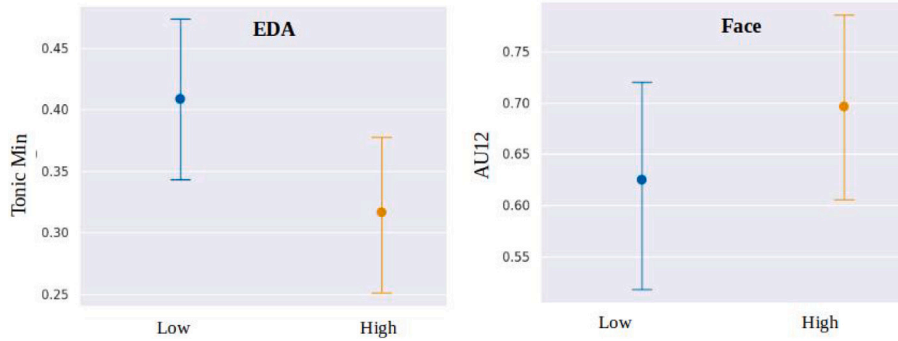


Fig. 4. The point-plot of features significantly affected by emotional arousal. AU10 and AU12 had similar plots.

7. Discussion

We opted for a quantitative approach to address our research questions and explore human body responses in face-to-face vs. remote conversational settings. The research questions necessitate a quantitative framework as they measure and compare specific physiological and behavioral cues, such as heart rate variability (HRV) and facial expressions, between face-to-face and remote conversational settings. A quantitative approach allows us to collect precise numerical data, facilitating statistical analyses that can uncover subtle differences and patterns. We also minimized subjective interpretation by employing

standardized data collection instruments and using common feature extraction and analysis methods. Moreover, this quantitative approach facilitates the extraction of generalizable insights that can inform our understanding of human responses in both face-to-face and remote conversational contexts.

Our research employed a rigorous methodology to ensure the accuracy and consistency of the results. We used a set of images from the IAPS dataset as the topic of conversation to minimize the effect of variety, imbalance, and complexity in induced emotions. Counterbalancing was implemented to control the effect of topic order and remote or face-to-face order. We considered a sample size that is in

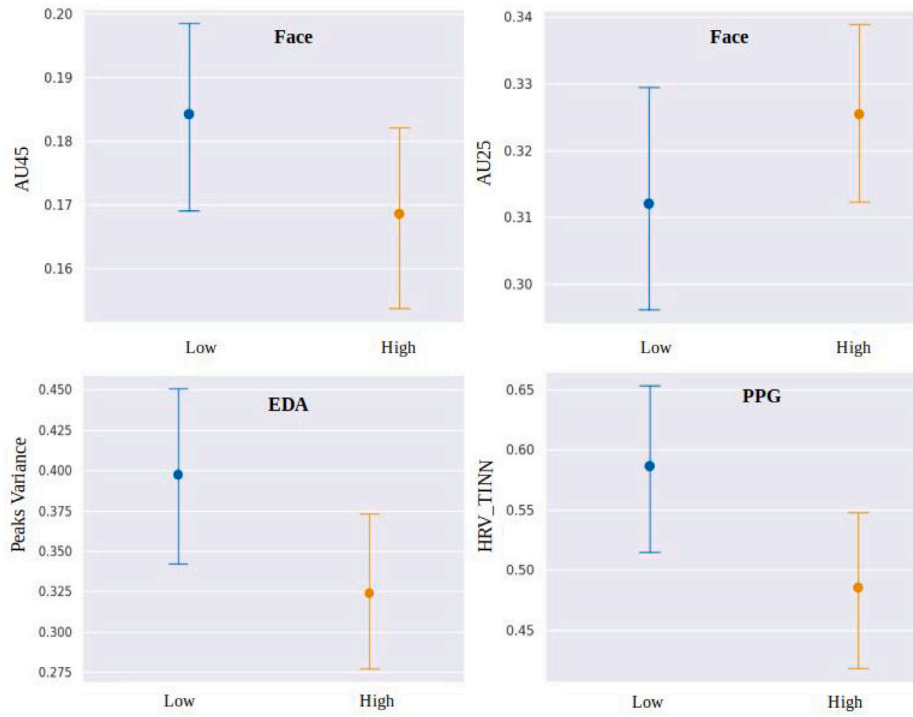


Fig. 5. The point-plot of features that are significantly affected by emotional valence. For the EDA signal, peaks' positive derivatives, peaks' frequency, and peaks' variance were correlated and had similar plots. For the face, AU06, AU10, and AU12 plots were similar to AU25, and AU04 was similar to AU45.

Table 6

The accuracy of recognizing arousal and valence using different modalities in face-to-face and remote conditions.

Arousal		kfold	LOSO	Pre-trained	Trained
EDA	f2f	75.8	57.7	92.4	66.8
	remote	76.9	54.9	99.8	62.9
PPG	f2f	75.9	56.9	88.1	66.5
	remote	73.8	55.6	99.2	62.3
Face	f2f	78.2	68.8	88.8	70.0
	remote	75.3	64.2	99.4	65.9

Valence		kfold	LOSO	Pre-trained	Trained
EDA	f2f	74.1	61.6	92.0	61.6
	remote	64.6	48.7	99.8	57.2
PPG	f2f	70.7	53.6	89.9	46.6
	remote	77.5	51.6	99.8	55.4
Face	f2f	68.2	54.6	94.7	58.3
	remote	74.7	54.2	100	55.6

the range of common multimodal datasets with physiological data for emotion recognition (see Table 1). We also used standard and commonly used devices for data collection and employed common methods for extracting features and data analysis. Moreover, we used Repeated measures ANOVA with ART to reduce the impact of non-normality and heteroscedasticity in data. We used multimodal measurements, and the results of analyzing various modalities were aligned together. For emotion recognition, we used a leave-one-subject-out (LOSO) or leave-one-out cross-validation strategy, which evaluates the model on unseen participants in the dataset. The LOSO strategy helped us assess the generalization performance of the model by testing it on data that it has not been trained on. It also helped us efficiently use available data by maximizing the data used for training in each iteration while still providing a reliable performance estimate.

Based on our result, people exhibited more facial expressions associated with positive emotions, such as AU45, AU06, AU12, and AU14,

during remote conversations. We also found that the average of the phasic component, which is associated with rapid changes in EDA, was significantly higher in face-to-face conversations than in remote conversations. Additionally, we found that people have higher heart rate variability (HRV) during remote conversations. These changes are associated with improved emotional well-being [50], decreased stress levels, reduced anxiety [51], and enhanced regulation of emotional reactions [52] in remote conversations vs. face-to-face conversations. Overall, our results show electrodermal activities, heart rate variability, and facial expressions differ in face-to-face and remote conditions. This suggests that people experience more positive emotions with higher arousal levels in remote conversations than in face-to-face conversations. This result can be because some people reported feeling more uncomfortable in face-to-face conversations with a stranger than in remote conversations. This is likely because face-to-face conversations can be more stressful, especially when people are talking to strangers for the first time. One participant expressed he felt more comfortable in remote settings because of a decreased sense of presence, so he was less nervous talking to a stranger. In addition, face-to-face conversations can be more difficult to escape if they become uncomfortable. In contrast, remote conversations are often more relaxed, especially when people are used to having remote conversations with strangers in everyday meetings related to their studies or work.

On the other hand, based on our result, people felt less empathy in remote conversations vs. face-to-face conversations. The human body responds differently in various conversational conditions for many individually specific reasons. One important reason is the lack of social presence because of missing non-verbal cues in remote conversation, like full-body posture and talking to a screen. This lack of social presence can make reading each other's emotions and intentions significantly more difficult [55]. Another reason can be experiencing a different environment than the conversational partner and experiencing various environmental stimuli. Face-to-face conversations can also occur in a more personal and intimate setting, making people feel more vulnerable and emotionally invested in the conversation. Remote conversations, on the other hand, often take place in a more impersonal

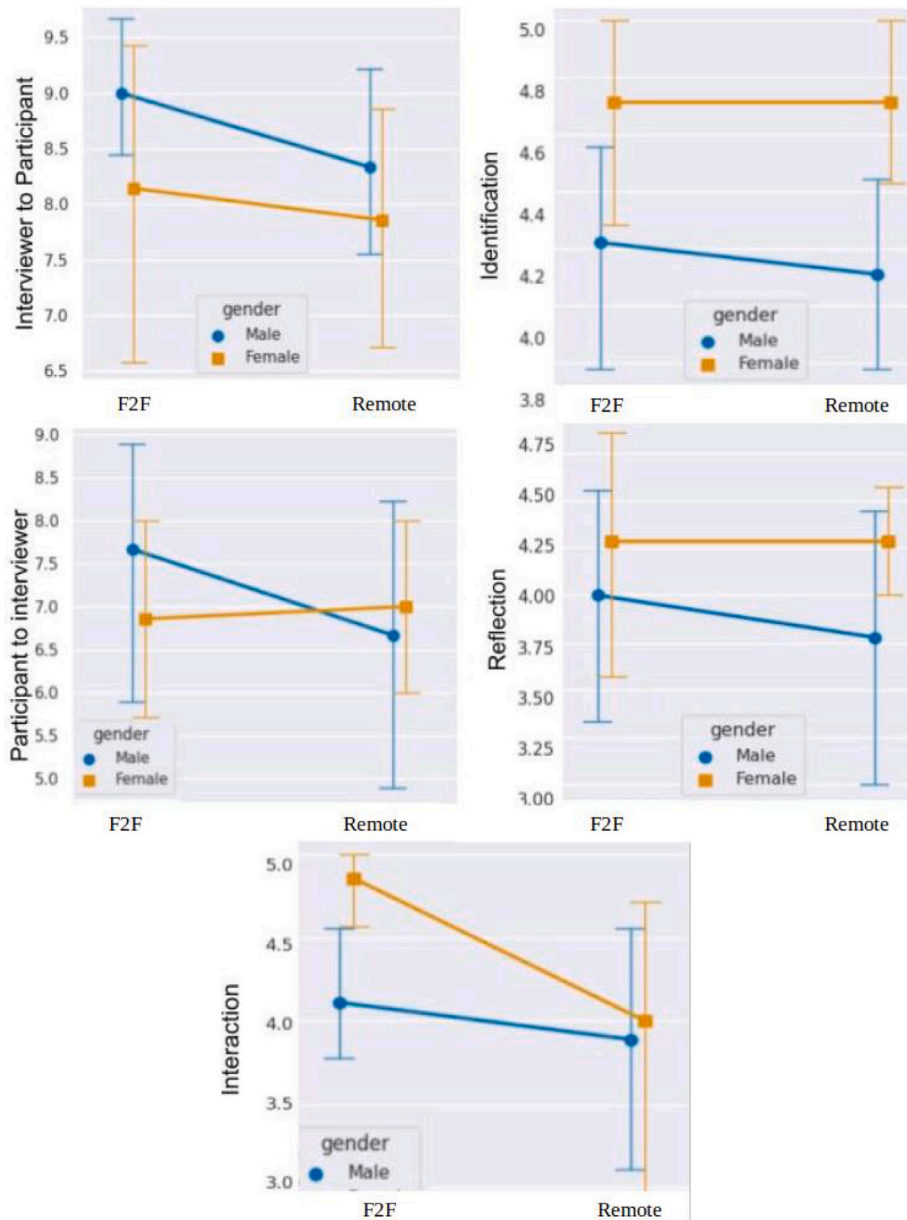


Fig. 6. The point plots for displaying the interaction between the gender and the condition (face-to-face or remote) on empathy factors and quality of interaction .

and detached setting, which can make people feel less emotionally involved. In remote conditions, there is also a lack of spatial cues because a person on a screen appears flat compared to someone in real life. This makes it much more challenging to interpret spatial cues.

In this paper, we were keen to know if any difference exists between human physiological responses in face-to-face and remote communications in emotional contexts. As discussed, our findings showed some features of heart rate activity, electrodermal activity, and facial action units are significantly different in face-to-face and remote conversations. This variety of results reveals the importance of research on improving remote interaction. This improvement can be done by detecting emotions using multimodal data in real-time and transferring it to the remote person as an additional cue for better interaction. We also investigated various empathy factors in different conversational conditions. We found that people were more empathetic in face-to-face conditions, possibly because of increased social presence and experiencing the same environment. For example, one of the participants mentioned, “I feel more connected to the other person when I see her sitting close to me”.

We employed a leave-one-subject-out (LOSO) strategy to evaluate the model’s accuracy, ensuring that the model generalizes well to new, unseen individuals. The final result is derived from the average accuracy of the model across all participants when each one is treated as the subject left out during testing. The results also show that training an emotion recognition model on one condition and testing it on another condition performs significantly better than a model trained and tested on the same condition. This suggests that there are similarities in how people’s bodies respond to different emotions, even in other conversational conditions. We found that pre-training the model with one condition and training it on another condition leads to considerably high accuracy in detecting emotions in the LOSO strategy. This high accuracy is because the model learned about the people’s responses in both conditions and the test person’s responses in another condition.

While our study design aimed to find differences in human body responses in face-to-face and remote conversations, it is essential to acknowledge the limitations of our work and issues that should be addressed before generalizing these findings. These include:

Sample Size: In this research, we aimed for a sample size of between 15 and 32 participants, as this aligns with the common range observed in well-known multimodal datasets involving physiological data for emotion recognition (Table 1). Achieving a larger sample size also proved difficult due to the inherent challenge of collecting both behavioral and physiological responses in both remote and face-to-face settings, requiring longer participation times, and the additional limitations imposed by COVID-19 restrictions. However, there are lots of variables, such as personality, ethnicity, and gender, which could directly affect the results. This dataset could be extended in the future by recruiting more participants and considering various factors mentioned in this section. Increasing the dataset size could decrease the effect of variety in people's data and outliers to make a more reliable and generalizable model.

Lack of EEG data: The EEG signals showed an improvement in multimodal emotion recognition. However, we missed this valuable source for emotion recognition because of sensor failure. Using a new EEG cap and more reliable data collection, we could consider EEG signals in emotion recognition and explore differences in brain activity in various settings.

Psychological Effect: People may respond differently and feel different emotions when discussing the same topic. To minimize this effect, we considered more than one topic for each conversation condition and emotional arousal and valence states and then calculated the results by averaging. However, only a few topics were discussed, so it is difficult to know if the results could be generalized to all conversation topics. In the future, psychological effects could be decreased by increasing the number of trials and conversation topics.

Background Skills: All participants were university staff and students who were quite familiar with Zoom calls and having remote communications in their daily work. So, our result may only apply to people familiar with video conferencing. In the future, these findings could be generalized by recruiting participants with different backgrounds and skills.

Interviewer Effect: Due to our limitations in recruiting participants, we did not consider the effect of the interviewer's gender on the result. Some people interact more easily with females, while others are more comfortable with males [56,57]. In the future, increasing the sample size will reduce the interviewer effect. Another solution could be repeating the study with a male interviewer to consider the impact of the interviewer's gender. In this study, all participants also had conversations with the same interviewer, so recruiting more interviewers would also minimize the effect of the interviewer's personality.

Familiarity Effect: In this study, the interviewer was a stranger to all participants. However, people may behave differently depending on how well they know the person they are talking to. Therefore, it would be interesting to conduct a future experiment where participants talk to their partners or close friends instead of strangers to see if the results differ.

8. Conclusions

This research aimed to investigate the differences in human behavioral and physiological responses between face-to-face and remote conversations in an emotional context. In this study, we addressed our research questions by creating a dataset of behavioral and physiological responses in both conversational settings and statistically explored the varieties of body responses and subjective empathy based on the type of conversational settings and emotional arousal and valence. We also trained a random forest classifier with each setting's data and tested it with the other setting's data to address the third research question. Our investigation revealed several key findings that offer meaningful insights into human-human and human-computer interactions. Based on our result, people exhibited more facial expressions associated with positive emotions, such as AU45, AU06, AU12, and AU14, during remote conversations. We also found that the average of

Table A.7

List of images from the IAPS dataset used as the conversation topic.

Image ID	Subject	Valence	Arousal	Label
8496	WaterSlide	7.58	5.79	HVHA
5833	Beach	8.22	5.71	HVHA
8185	Skydivers	7.57	7.27	HVHA
8380	Athletes	7.56	5.74	HVHA
1441	PolarBears	7.97	3.94	HVLA
1630	Fawn	7.26	4.45	HVLA
5760	Nature	8.05	3.22	HVLA
5764	Field	6.74	3.55	HVLA
1201	Spider	3.55	6.36	LVHA
9910	CarAccident	2.06	6.20	LVHA
9927	Flood	2.71	5.29	LVHA
9075	StarvingChild	1.66	6.04	LVHA
9913	Truck sticking in mud	4.38	4.42	LVLA
9220	Cemetery	2.06	4.00	LVLA
2205	Hospital	1.95	4.53	LVLA
9280	Smoke, pollution	2.80	4.26	LVLA

the phasic component, which is associated with rapid changes in EDA, was significantly higher in face-to-face conversations than in remote conversations. Additionally, we found that people have higher heart rate variability during remote conversations. Finally, we showed that although differences were observed, there was still enough commonality to support the development of generalizable models. We showed an emotion recognition model can be pre-trained with one condition's data to improve the emotion recognition accuracy in the other condition.

One of the major theoretical contributions of this study lies in its exploration of the effect of remote interactions on the human body. Using a quantitative approach and a set of statistical analyses, we have extended existing knowledge by quantitatively validating how the human body responds differently in face-to-face and remote emotional conversations. We also discussed how these varieties could be due to a lack of social presence, experiencing various environments, missing some non-verbal cues, and experiencing a detached and impersonal setting in remote conversations. Our findings also showed that despite these differences, there is still the possibility of developing generalizable emotion recognition models that can be used in various conversational settings. This nuanced understanding of human body responses in various settings is vital for designing more effective ways for people to communicate and be connected, even when they are not physically present in the same space.

The experimental study design and the created multimodal dataset in this research serve as a foundational basis for further investigations, offering a more comprehensive understanding of the effect of remote interactions on human body responses. This multimodal dataset could be used to design more complex and advanced models for emotion recognition that can be used for recognizing emotions in various real-world applications.

In identifying and addressing gaps in the current literature, this study has laid a foundation for future research endeavors. By elucidating the unexplored territory of how remote interaction affects human body responses, we have highlighted avenues for further investigation to advance the designing of more effective ways for human-human and human-machine interactions. We recommend that future research should consider the effect of significantly different features identified in this research in designing systems with emotion recognition capabilities. We can justify the quality of remote conversations by monitoring significant features of each signal. Also, relying on these research findings, we could design tools that can adapt to user emotions in real-time, providing feedback or adjusting communication styles for more effective engagement.

Our reliance on facial expression, EDA, and PPG analysis might miss other important nonverbal cues in remote communication, so in the future, we suggest including more channels, such as eye gaze, body gestures, and brain activity, to better understand the effect of remote

communications on the human body. Our findings provide valuable insights into how the human body responds differently in face-to-face vs. remote conversations. However, the specific focus on frequent video conference users and limited sample size restrict applying these findings to broader populations or cultural settings. Future research with more diverse samples is needed to generalize these findings. Moreover, while our controlled lab experiment provided valuable insights, it may not fully capture the complexities of real-world remote interactions. Thus, identifying the impact of different video conference platforms or communication technologies on human responses and perceived social presence could be an important future research direction.

CRedit authorship contribution statement

Nastaran Saffaryazdi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nikita Kirkcaldy:** Data curation. **Gun Lee:** Formal analysis. **Kate Loveys:** Investigation. **Elizabeth Broadbent:** Writing – review & editing, Supervision, Conceptualization. **Mark Billingham:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Selected IAPS images

Table A.7 shows the selected images from the IAPS dataset and their subjects, along with their arousal and valence scores in the dataset.

Appendix B. Interviewer's questions

The detailed questions that the interviewer used to have the conversation are as follows.

- *What emotion did you feel when you watched the image? Why?*
- *Tell me about a time when you experienced something similar to the image.*
- *What was going on for you then?*
- *Tell me about how you felt that experience in your senses (giving them an example of feeling hot when you're angry or starting to shake when you're nervous/anxious)*
- *If you close your eyes now and put yourself in the image scene, can you tell me what is going on for you (senses, imagination, etc.*
- *How do you feel reflecting on that moment now?*

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